# Term Project Guidelines

EE5203 Microprocessors — 2026-2027 Fall

Basri Erdogan

## 1 What You Will Build

In a team of **2 or 3 students**, you will build an embedded system on the **ST Nucleo-F401RE** board. The system reads data from at least one sensor, processes it, shows or uses the result, and communicates over a serial bus. You develop it step by step through milestones that are checked in your lab sessions, and you finish with a demonstration and an individual oral defence. The project is **20%** of your course grade.

## 2 Key Dates

Every team has a **home section**: the lab section you name when you register (Lab 1: Tuesday 13:00, Lab 2: Tuesday 15:00, Lab 3: Wednesday 13:00). Your milestones are checked in the lab of your home section.

| Week | Milestone | What you show | Date |
|---|---|---|---|
| 5 | Team registration | E-mail your team members (names, student numbers, lab sections), your home section and your topic to the milestone assistant of your home section | **18 October 2026, 23:59** |
| 7 | Proposal | One-page PDF on CATS: problem, block diagram, components, minimum working system (Section 4.1) | Before the theory lecture, **Monday 26 October, 11:00** |
| 9 | Design review and first hardware | Design document on CATS before the lab: pin and peripheral table, data flow, who does what, risks. In the lab: one input or output part working | Lab, **10/11 November** |
| 10 | Data acquisition | Sensor data read on the board and checked against a known value | Lab, **17/18 November** |
| 11 | Communication | UART log or command interface (or your I2C/SPI device) working | Lab, **24/25 November** |
| 12 | Submission | Report, slides and source code on CATS | **29 November 2026, 23:59** |
| 12 | Demonstration and oral defence | Live demonstration, presentation and individual defence, in one joint session for all teams in our laboratory room | **Tuesday 1 December, from 13:00** |

There is no make-up for a missed milestone or the demonstration, and late submissions are not accepted.

**Milestone assistants.** The milestone assistant of your home section registers your team, receives your uploads and checks your milestones:

- Lab 1 and Lab 2: Res. Asst. Ahmet Aytuğ Ayrancı (ahmet.ayranci@iku.edu.tr)
- Lab 3: Res. Asst. Ahmet Can Tanık (a.tanik@iku.edu.tr)

**How the checks in weeks 9, 10 and 11 work.** The check takes the first part of the lab of your home section. Right after the opening brief, your milestone assistant calls the teams one at a time, in team-number order, for a 3-minute check: you show the milestone working on your hardware. At least one team member must be there with the hardware; members from other sections attend their own lab as usual. Have your hardware connected and your code flashed before your team is called. A team that is not ready when called misses the milestone. The result is pass or miss. The rest of the lab runs as usual.

# 3 Rules for Teams and Topics

- A team has **2 or 3 members**. Members may come from different lab sections; the team names one **home section** when it registers.
- Choose a topic from Section 9 or propose your own. Your proposal (due 26 October) must be approved before you continue.
- No two teams in the course can build the same project. The first team to register a topic gets it.
- After the proposal you cannot change your team or topic without the instructor's approval.
- **Parts:** the course kit has no sensors. Your team buys the sensors, displays and actuators your project needs. Order them **before the proposal**; parts that arrive late do not move a milestone. Until a part arrives, you may use the potentiometer as a stand-in if you say so.

# 4 Technical Requirements

Your system **must** have all of these parts:

1. **Board and tools:** Nucleo-F401RE, programmed in C with STM32CubeMX and VS Code (HAL or register level).
2. **Data acquisition:** at least one sensor, read through the ADC or a digital interface.
3. **Processing:** a clear method that turns raw data into a decision or a result (for example averaging and thresholds, a state machine, a controller).
4. **Output:** show or use the result on a suitable device (display, LEDs, buzzer, servo or motor).
5. **Communication:** at least one serial protocol — UART, I2C or SPI — between the board and a sensor, a display or a PC.

**Minimum complexity.** A sensor connected to one LED is not enough. Your system must use **at least three** of the course topics below, and its timing must not depend on `HAL_Delay()` in the main loop:

| Course topic | Lecture |
|---|---|
| External interrupts (EXTI) | L06 |
| Timers and non-blocking timing | L07 |
| PWM, output compare or input capture | L08 |
| ADC | L09 |
| UART | L10 |
| I2C or SPI | L11 |

## 4.1 Minimum working system

Your **minimum working system** is the smallest version of your project that you promise will work at the demonstration on 1 December. If everything else fails, this part must still run. In your proposal, name its five parts:

1. **Sensor:** which one, and how you read it (ADC, I2C, …).
2. **Processing:** what your code does with the data (average, threshold, state machine, …).
3. **Output:** what shows or uses the result (LED, display, buzzer, motor, …).
4. **Serial link:** UART, I2C or SPI, and between which devices.
5. **Course topics:** the three or more topics from the table above that it uses.

The minimum working system alone must meet all five requirements above and the minimum complexity rule. List any other features separately, under "If time allows". They are not promised.

# 5 What You Submit

Upload **one ZIP file per team** to CATS by 29 November, 23:59. Name it `EE5203_Project_TeamNN.zip`. It contains:

1. **Source code:** the CubeMX project (`.ioc` file and the `Core` folder). Do not include the `build` folder.
2. **Report (PDF, 10-15 pages)** using the report template on CATS. It must contain:
   - the **requirements table**: each requirement, how you test it, and the measured result;
   - the **resource map**: every pin, timer, interrupt and peripheral you use, and what it does;
   - the **timing budget**: how often each task runs and how long it takes;
   - design decisions, wiring diagram, evidence (photos, logs, measurements) and known issues;

- **work distribution**: which member wrote which part;
- **AI use**: which tools you used and for what (see Section 7). Write "none" if you used none.

3. **Slides (PDF): 6 to 8 slides per team, not counting the title slide.** Every member presents at least 2 of them.

Keep a working copy of your final code that you can flash in under two minutes on the day.

# 6 Demonstration and Oral Defence

All teams demonstrate in **one joint session on Tuesday 1 December, from 13:00, in our laboratory room**. The running order is announced on CATS. Every member must be present for the whole session.

**15 minutes per team in total, presentation, demonstration and questions included.** Set up while the team before you presents: your system must already be running when your slot starts. A slot that runs over is stopped.

| Time | What you do |
|---|---|
| 0-8 min | Presentation with live demonstration: every member presents at least 2 slides; show each requirement on the running system as you present it (one requirement, one test, one observed result) |
| 8-9 min | **One fault you inject on purpose** (say first what you expect) |
| 9-15 min | Individual questions: about 3 minutes per member in a team of two, 2 minutes in a team of three |

Your known issues belong in the report; you may be asked about them.

**Individual questions.** Each member answers short questions about their own contribution, at up to three levels:

1. **Your contribution:** "Which parts did you write? Walk me through one."
2. **The mechanism:** "Which register does that configure? What does the hardware do with it?"
3. **The alternative:** "Why this way? What would happen if you did it differently?"

Be ready to open the code and the register view and point to the answer.

# 7 Use of AI Tools

These rules come from the course syllabus.

- **Allowed:** asking an AI tool to explain a concept, a compiler or debugger message, or a datasheet passage; help with report wording; review of code you wrote yourself.
- **Not allowed:** generating project code; producing the analysis or conclusions of the report with AI.
- **Disclosure:** state in the report how you used AI tools.
- **Check:** the oral defence. Work you cannot explain and change yourself earns no credit.

# 8 Grading

| Part | Points | Graded per |
|---|---|---|
| Demonstration against your requirements | 35 | team |
| Report | 25 | team |
| Presentation (the slides you present) | 15 | member |
| Individual oral defence | 25 | member |

**Milestones.** Each milestone in weeks 6, 9, 10 and 11 that is missed costs the team **5 points**. A milestone is missed when the upload is not on CATS when the lab of your home section starts, or when the check in that lab does not pass.

**Team points** = demonstration + report (at most 60). **Your share of the team points depends on your defence:**

**Your project grade = (team points – milestone deductions) × defence multiplier + your presentation points + your defence points**

| Your result in the oral defence | Defence multiplier |
|---|---|
| You explain your contribution **and** the hardware mechanism behind it | × 1 |
| You explain your contribution, but not the mechanism (register, peripheral) | × 0.5 |
| You cannot explain your own contribution | × 0 (project grade = 0) |

**Example.** A team scores 30 (demonstration) + 20 (report) = 50 team points and missed no milestone.

| Member | Presentation and defence | Calculation | Project grade |
|---|---|---|---|
| A | presentation 12; explains contribution and mechanism, defence 22 | 50 × 1 + 12 + 22 | 84 |
| B | presentation 10; explains contribution, not the mechanism, defence 12 | 50 × 0.5 + 10 + 12 | 47 |
| C | cannot explain own contribution | 0 | 0 |

# 9 Project Ideas

Each idea lists one possible set of parts and course topics, and how you can check your measurement against a known value. You may change the parts or propose your own idea.

| # | Project | Sensors and outputs | Course topics | Check against |
|---|---|---|---|---|
| 1 | Parking assistant | HC-SR04 ultrasonic sensor; buzzer; LED bar or display | input capture, PWM, timers, UART | tape measure |
| 2 | Room climate monitor with alarm | BME280 or SHT31 sensor (I2C); I2C OLED or LCD; buzzer | I2C, timers, EXTI (alarm reset), UART log | room thermometer |
| 3 | Line-following robot | IR line sensors; TB6612FNG motor driver; robot chassis | ADC or GPIO, PWM, timers, EXTI (start/stop), UART tuning log | sensor values on black and white |
| 4 | Tilt-stabilised camera platform | MPU6050 IMU; two hobby servos (two-axis gimbal) | I2C, PWM, timers (control loop), UART tuning | protractor angles |
| 5 | Sunrise alarm clock | DS3231 real-time clock; light sensor (LDR); LED that brightens before the alarm; display; buzzer | I2C, ADC, PWM, EXTI (buttons), timers | phone clock; covered and lit LDR |
| 6 | Motion alarm system | PIR sensor; keypad or buttons for the code; buzzer; display | EXTI, timers, PWM (siren), UART event log | walk-past tests at set distances |
| 7 | Vibration and tilt monitor | ADXL345 or MPU6050 accelerometer; vibration motor as a test source; buzzer | I2C or SPI, timers (fixed sampling rate), EXTI (data-ready pin), UART stream to a PC plot | 1 g when flat, 0 g on its side |
| 8 | Digital weighing scale | Load cell with HX711 amplifier; display; tare button | timers, EXTI (tare), I2C display, UART log | known weights (water bottle, coins) |
| 9 | Temperature-controlled fan | LM35 or NTC thermistor; DC fan with transistor; LCD | ADC, PWM (fan speed), timers, I2C display | room thermometer |
| 10 | Colour sorter | TCS34725 colour sensor; servo that sorts objects into bins | I2C, PWM (servo), timers, UART | sorting reference cards |
| 11 | Motor speed controller | DC motor with encoder or slotted speed sensor; motor driver | input capture (speed), PWM, timers (control loop), UART commands (set speed) | phone stroboscope or tachometer app |
| 12 | Ultrasonic radar scanner | HC-SR04 on a servo that sweeps 0-180°; PC plot of angle and distance | PWM, input capture, timers, UART to PC | objects at measured positions |

| # | Project | Sensors and outputs | Course topics | Check against |
|---|---|---|---|---|
| 13 | Pulse-rate monitor | MAX30102 (I2C) or analog optical pulse sensor; OLED display | ADC or I2C, timers (sampling), I2C display, UART log | pulse counted by hand for 30 s |
| 14 | Plant watering controller | Capacitive soil-moisture sensor; small pump with transistor; display | ADC, timers, PWM or GPIO (pump), UART commands | dry soil, wet soil, glass of water |

**Electrical rules.** The pins work at 3.3 V: a 5 V sensor output needs a voltage divider or a level shifter. Motors, pumps and fans need a driver with a flyback diode and their own supply, never a pin. No mains (230 V) wiring. The pulse-rate monitor is a demonstration, not a medical device.